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CABLE CONNECTOR ASSEMBLY HAVING LOCATING FEATURES

Abstract

A connector assembly includes a connector holder having a frame holding a cable connector in a connector chamber. The cable connector includes a connector housing holding contact assemblies. The connector assembly includes a mounting assembly for mounting the connector holder to a panel. The mounting assembly includes a shaft, a biasing member operably coupled between the connector holder and the shaft, and a support element configured to be operably coupled between the shaft and the panel to support the shaft relative to the panel. The shaft is received in a locating opening and is undersized relative to the locating opening to allow a limited amount of floating movement in the locating opening in a float direction perpendicular to the mating direction to allow the connector holder and the cable connector to move relative to the panel.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims benefit to U.S. Provisional Application No. 63/560,837, filed 4 Mar. 2024, titled "CABLE CONNECTOR ASSEMBLY HAVING LOCATING FEATURES", the subject matter of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] The subject matter herein relates generally to communication systems.

[0003] Communication systems use electrical connectors to transmit data between components of the communication system. For example, electrical connectors may be held in a rack, chassis, cabinet or other system component. Mating connectors are mated to the electrical connectors. For example, the mating connectors may be mounted to a card assembly, such as a daughtercard or server card, which is plugged into the chassis. Proper mating of the mating connector with the electrical connector is necessary for proper operation of the communication system. For example, misalignment of the connectors may lead to damage to the contacts of the connectors. Additionally, partial mating of the connectors may lead to reduction in signal integrity due to partial mating of contacts and/or improper shielding of the signal contacts. In some systems, the server card may be mated with the assistance of a lever or other mechanical assist device to plug the server card into the chassis. Such mechanical loading of the server card may lead to damage to the electrical connector. Additionally, mounting of the electrical connectors within the rack may require special tools or mounting hardware, which adds cost and complexity to the system.

BRIEF DESCRIPTION OF THE INVENTION

[0004] In one embodiment, a connector assembly is provided and includes a connector holder having a frame defining a connector chamber. The frame extends between a front and a rear of the connector holder. The frame includes an upper frame member, a lower frame member, and first and second side frame members extend between the upper and lower frame members to define the connector chamber. The connector assembly includes a cable connector received in the connector chamber. The cable connector includes a connector housing holding contact assemblies. Each contact assembly includes a signal contact and a cable terminated to the signal contact. The signal contact configured to be mated with a mating signal contact of a mating connector in a mating direction. The connector assembly includes a mounting assembly for mounting the connector holder to a panel. The mounting assembly includes a shaft, a biasing member operably coupled between the connector holder and the shaft, and a support element configured to be operably coupled between the shaft and the panel to support the shaft relative to the panel. The shaft configured to be received in a locating opening, wherein the shaft is undersized relative to the locating opening to allow a limited amount of floating movement in the locating opening in a float direction perpendicular to the mating direction to allow the connector holder and the cable connector to move relative to the panel. The biasing member forward biasing the cable connector and is compressible in a rearward compression direction parallel to the mating direction to allow the connector holder and the cable connector to move rearward relative to the panel.

[0005] In another embodiment, a connector assembly is provided and includes a connector holder having a frame defining a connector chamber. The frame extends between a front and a rear of the connector holder. The frame includes an upper frame member, a lower frame member, and first and second side frame members extend between the upper and lower frame members to define the connector chamber. The connector assembly includes a first cable connector received in the connector chamber. The first cable connector includes a first connector housing holding first contact assemblies. Each first contact assembly includes a first signal contact and a first cable terminated to the first signal contact. The first signal contact configured to be mated with a first mating signal contact of a first mating connector in a mating direction. The connector assembly includes a second cable connector received in the connector chamber. The second cable connector includes a second connector housing holding second contact assemblies. Each second contact assembly includes a second signal contact and a second cable terminated to the second signal contact. The second signal contact configured to be mated with a second mating signal contact of a second mating connector in a mating direction. The connector assembly includes a mounting assembly for mounting the connector holder to a panel. The mounting assembly includes a shaft, a biasing member operably coupled between the connector holder and the shaft, and a support element configured to be operably coupled between the shaft and the panel to support the shaft relative to the panel. The shaft configured to be received in a locating opening, wherein the shaft is undersized relative to the locating opening to allow a limited amount of floating movement in the locating opening in a float direction perpendicular to the mating direction to allow the connector holder and the cable connector to move relative to the panel. The biasing member forward biasing the cable connector and is compressible in a rearward compression direction parallel to the mating direction to allow the connector holder and the cable connector to move rearward relative to the panel.

[0006] In a further embodiment, a communication system is provided and includes a cartridge having panels forming a chamber. The panels include a front panel. The cartridge includes ports in the front panel at a front of the cartridge. The communication system includes connector assemblies received in the ports for mating with mating connector assemblies. Each connector assembly includes a connector holder, a cable connector held by the connector holder, and a mounting assembly for mounting the connector holder and the cable connector to the cartridge. Each connector holder has a frame defining a connector chamber. The frame extends between a front and a rear of the connector holder. The frame includes an upper frame member, a lower frame member, and first and second side frame members extend between the upper and lower frame members to define the connector chamber. Each cable connector is received in the corresponding connector chamber. Each cable connector includes a connector housing holding contact assemblies. Each contact assembly includes a signal contact and a cable terminated to the signal contact. The signal contact configured to be mated with a mating signal contact of a mating connector in a mating direction. Each mounting assembly includes a shaft, a biasing member operably coupled between the connector holder and the shaft, and a support element configured to be operably coupled between the shaft and the panels of the cartridge to support the shaft relative to the panels. The shaft configured to be received in a locating opening, wherein the shaft is undersized relative to the locating opening to allow a limited amount of floating movement in the locating opening in a float direction perpendicular to the mating direction to allow the connector holder and the cable connector to move relative to the panels. The biasing member forward biasing the cable connector and is compressible in a rearward compression direction parallel to the mating direction to allow the connector holder and the cable connector to move rearward relative to the panels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a front view of a communication system in accordance with an exemplary embodiment.

[0008] FIG. **2** is a front view of a portion of the communication system in accordance with an exemplary embodiment.

[0009] FIG. **3** illustrates a portion of the communication system showing two of the cartridges in accordance with an exemplary embodiment.

[0010] FIG. **4** is a perspective view of a portion of the communication system showing one of the mating connector assemblies poised for coupling to the corresponding connector assembly in accordance with an exemplary embodiment.

[0011] FIG. **5** is a front perspective view of a portion of the communication system showing one of the connector assemblies in accordance with an exemplary embodiment coupled to the front panel of the cartridge.

[0012] FIG. **6** is a front perspective, exploded view of the connector assembly in accordance with an exemplary embodiment.

[0013] FIG. **7** is a front perspective view of a portion of the communication system showing one of the connector assemblies in accordance with an exemplary embodiment coupled to the front panel of the cartridge.

[0014] FIG. **8** is a front perspective, exploded view of the connector assembly in accordance with an exemplary embodiment.

[0015] FIG. **9** is a cross-sectional view of a portion of the communication system showing the connector assembly mounted to the front panel of the cartridge in accordance with an exemplary embodiment.

[0016] FIG. **10** is a perspective view of a portion of the communication system showing one of the connector assemblies in accordance with an exemplary embodiment.

[0017] FIG. **11** is a front perspective view of a portion of the communication system showing one of the connector assemblies in accordance with an exemplary embodiment.

[0018] FIG. **12** is a front perspective, exploded view of a portion of the connector assembly showing the cable connectors and the connector holder in accordance with an exemplary embodiment.

[0019] FIG. **13** is a front perspective, exploded view of a portion of the connector assembly showing the mounting assembly in accordance with an exemplary embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0020] FIG. **1** is a front view of a communication system **100** in accordance with an exemplary embodiment. FIG. **2** is a front view of a portion of the communication system **100** in accordance with an exemplary embodiment. In an exemplary embodiment, the communication system **100** includes a chassis or rack **50** having a rack frame **52** holding electrical components **60**. The rack **50** may be a server rack. The electrical components **60** may include fixed modules **62** (FIG. **2**) and/or removable modules **64** (FIG. **1**). The removable modules **64** may be mated to the fixed modules **62**. For example, the removable modules **64** may be plugged into the rack frame **52** and removable from the rack frame **52**. In various embodiments, the electrical components **60** may include one or more patch panels, switches, servers, routers, firewalls, and the like. The rack frame **52** may hold a power supply, cooling components, and the like.

[0021] In an exemplary embodiment, the communication system **100** includes at least one connector assembly **102** (FIG. **2**) and at least one mating connector assembly **104** (FIG. **4**) configured to be mated with the corresponding at least one connector assembly **102**. The connector assembly **102** may be part of one of the electrical components **60**, such as one of the fixed modules **62**. The mating connector assembly **104** may be part of one of the electrical components **60**, such as one of the removable modules **64**. In various embodiments, each mating connector assembly **104** may include one or more mating electrical connectors mounted to a circuit board, such as a backplane, a daughter card, a network switch, and the like.

[0022] In an exemplary embodiment, the communication system **100** includes a cabinet or cartridge **110** that holds the connector assemblies **102**. The connector assemblies **102** may be cable connector assemblies having cables extending from corresponding cable connectors. The cartridge **110** may form part of the fixed modules **62**. Optionally, multiple cartridges **110** may be provided (for example, left side and right side). The cartridge **110** may be coupled to the rack frame **52**, such as at the rear of the rack frame **52**. The cartridge **110** forms an enclosed space or chamber **111** for the connector assemblies **102** and the cables of the connector assemblies **102**. The cartridge **110** is used for cable management. For example, the cables may be routed within the enclosed chamber **111** formed by the cartridge **110** to electrically connect between the various connector assemblies **102**. The cartridge **110** is used to support or position the connector assemblies **102** for mating with the mating connector assemblies **104**.

[0023] FIG. **3** illustrates a portion of the communication system **100** showing two of the cartridges **110** in accordance with an exemplary embodiment. Each cartridge **110** holds a plurality of the connector assemblies **102**, such as in one or more rows and one or more columns. The connector assemblies **102** are provided at the front of the cartridge **110** for interfacing with the mating connector assemblies **104** (shown in FIG. **4**).

[0024] In an exemplary embodiment, the cartridge **110** is formed from a plurality of panels **112**, such as sheet metal panels. The cartridge **110** includes a front panel **114**, side panels **116**, and a rear panel **118**. The panels **112** may additionally include an upper panel and/or a lower panel. The connector assemblies **102** are provided at the front panel **114** for mating with the mating connector assemblies **104**. In an exemplary embodiment, the connector assemblies **102** are arranged in a column along the front panel **114**. Optionally, the connector assemblies **102** may be provided in multiple columns. The mating ends of the connector assemblies **102** are external to the cartridge **110**. For example, the connector assemblies **102** pass through ports or openings **120** in the front panel **114**. The connector assemblies **102** are coupled to the front panel **114** adjacent the openings **120**. In an exemplary embodiment, the connector assemblies **102** may float or move relative to the front panel **114** to align with and mate to the mating connector assemblies **104**. In an exemplary embodiment, the connector assemblies **102** include mating guides, such as guideposts **122**, to guide mating of the mating electrical connectors **104** with the connector assemblies **102**. The guideposts **122** extend forward of the front panel **114**.

[0025] FIG. **4** is a perspective view of a portion of the communication system **100** showing one of the mating connector assemblies **104** poised for coupling to the corresponding connector assembly **102**. The connector assembly **102** is coupled to the cartridge **110**, such as to the front panel **114**. In the illustrated embodiment, the mating connector assembly **104** is a network component. For example, the mating connector assembly **104** may be a compute node, a memory module, a network switch, a storage device, a network accelerator, or another type of communication component. The mating connector assembly **104** may be a backplane component or a daughtercard component.

[0026] In an exemplary embodiment, the mating connector assembly **104** includes a mating electrical connector **106** mounted to a circuit board **108**. The mating connector assembly **104** may be mounted to a front edge of the circuit board **108**. The mating electrical connector **106** includes a connector housing **190** holding signal contacts (not shown). The signal contacts are terminated to the circuit board **108**. The signal contacts may be socket contacts. The mating electrical connector **106** may include shields providing shielding for the signal contacts.

[0027] The connector assembly **102** is received in the cartridge **110**. The connector assembly **102** is provided at the front panel **114** for mating with the mating connector assembly **104** along a mating axis **126**. In the illustrated embodiment, the mating axis **126** extends along the Z-axis. The connector assemblies **102**, **104** are configured to transmit and/or receive data through an interface. The connector assemblies **102** may includes receptacle connectors and the mating connector assemblies **104** may include plug connectors. In an exemplary embodiment, the guidepost **122** is configured to be mated with a guide module **124** of the mating connector assembly **104** to guide mating of the mating connector assembly **104** with the connector assembly **102**. In the illustrated embodiment, the guide module **124** includes an opening that receives the guidepost **122**. The mating guides may be used to align the connectors in one or more lateral directions transverse to the mating axis **126**, such as along the X-axis and/or the Y-axis. In various embodiments, the

mating guides may provide both horizontal alignment (X-axis) and vertical alignment (Y-axis).
[0028] FIG. **5** is a front perspective view of a portion of the communication system **100** showing one of the connector assemblies **102** in accordance with an exemplary embodiment coupled to the front panel **114** of the cartridge **110**. FIG. **6** is a front perspective, exploded view of the connector assembly **102** in accordance with an exemplary embodiment. The connector assembly **102** includes one or more cable connectors **130**, a connector holder **200** holding the cable connector(s) **130**, and a mounting assembly **300** for mounting the connector holder **200** and the cable connector(s) **130** to the front panel **114** of the cartridge **110**.
[0029] In the illustrated embodiment, the connector assembly **102** includes two cable connectors **130** arranged side-by-side. However, greater or fewer cable connectors **130** may be provided in alternative embodiments. The cable connectors **130** may be vertically stacked rather than horizontally stacked in alternative embodiments. The mating ends of the cable connectors **130** pass through the opening **120** in the front panel **114** for mating with the mating electrical connector **106** (FIG. **4**).
[0030] Each cable connector **130** includes a connector housing **132** holding contact assemblies **140**. The connector housing **132** includes a cavity **134** that receives the mating end of the mating electrical connector **106**. The contact assemblies **140** are arranged in the cavity **134**, such as in rows and columns. The walls of the connector housing **132** may be chamfered and have a lead-in surfaces to guide mating of the mating electrical connector **106** in the cavity **134**. The connector housing **132** may have guide features to properly position the mating electrical connector **106** within the cavity **134**.
[0031] Each contact assembly **140** includes at least one signal contact **142**, which is configured to be electrically connected to a wire or conductor of the corresponding cable **150** extending from the cable connector **130**. In the illustrated embodiment, each contact assembly **140** includes a pair of the signal contacts **142**, which define a differential pair. In various embodiments, the signal contacts **142** may be pin contacts. However, in alternative embodiments, the signal contacts **142** may be socket contacts, spring beam contacts, or other types of contacts. The signal contacts **142** may be stamped and formed contacts. Each contact assembly **140** includes a shield **144** provide electrical shielding for the signal contacts **142**. The shield **144** is configured to be electrically connected to the wire shield of the wire and/or the cable shield of the cable. In the illustrated embodiment, the shield **144** is C-shaped providing electrical shielding on three sides of the pair of signal contacts **142**. The shield **144** may have other shapes in alternative embodiments. The shield **144** may be a stamped and formed shield.
[0032] The connector holder **200** is used to hold the cable connector(s) **130** relative to the front panel **114**. The connector holder **200** is configured to be coupled to the cartridge **110**, via the mounting assembly **300**, to position each cable connector **130** relative to the cartridge **110**. In an exemplary embodiment, the cable connector **130** is movable relative to the connector holder **200** to position the cable connector **130** during mating with the mating electrical connector **106** (FIG. **4**). For example, the connector holder **200** and the cable connector **130** may be movable along the mating axis **126** (along the Z-axis) to accommodate overtravel of the mating connector assembly **104** during mating. In various embodiments, the connector holder **200** and the cable connector **130** may have a limited amount of floating movement relative to the front panel **114** to accommodate misalignment of the cable connector **130** relative to the mating electrical connector **106**. For example, the mounting assembly **300** allows movement of the connector holder **200** and the cable connector **130** relative to the front panel **114** along the X-axis and/or along the Y-axis to accommodate misalignment of the cable connector **130** relative to the mating electrical connector **106**. The mounting assembly **300** may limit or confine the amount of floating movement in the lateral floating directions X/Y. For example, the mounting assembly **300** may limit floating movement, such as to approximately 5.0 mm for mating tolerance. In various embodiments, the mounting assembly **300** may limit floating movement to approximately 3.0 mm for mating

tolerance.

[0033] The connector holder **200** includes a frame **202** defining a connector chamber **204**. The cable connector(s) **130** are received in the connector chamber **204**. The frame **202** supports the cable connector **130** in the connector chamber **204**. The frame **202** extends between a front **206** and a rear **208** of the connector holder **200**. The mating ends of each cable connector **130** is provided at or forward of the front **206** of the connector holder **200** for mating with the mating electrical connector **106**. The cables **150** extends from the rear **208** of the connector holder **200**. The guideposts **122** extend forward from the front **206**, such as above the cable connector **130**. Other locations are possible in alternative embodiments, such as the bottom for one or both sides of the frame **202**.

[0034] In an exemplary embodiment, the frame **202** includes an upper frame member **210**, a lower frame member **212**, a first side frame member **214**, and a second side frame member **216**. The connector chamber **204** is defined between the upper frame member **210** and the lower frame member **212**. The connector chamber **204** is defined between the first and second side frame members **214**, **216**. The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be plates or blocks. For example, the upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be stamped and formed plates. The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be die cast or molded blocks. In an exemplary embodiment, the upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be discrete pieces secured together using fasteners, clips, latches, welding or other securing means. In other various embodiments, the upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be integral with each other. The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and the second side frame member **216** form a brick **220** used to hold the cable connector **130**. The brick **220** is a rectangular structure that surrounds the connector chamber **204**. The brick **220** is configured to be coupled to the cartridge **110** to hold the cable connector **130** relative to the cartridge **110**.

[0035] In an exemplary embodiment, the frame **202** includes mounting features **224** configured to be coupled to the mounting assembly **300**. In the illustrated embodiment, the mounting features **224** include openings **226** in the frame members, such as in the upper frame member **210** and/or the lower frame member **212**. In the illustrated embodiment, four openings **226** are provided near the four corners of the brick **220**. In an exemplary embodiment, the mounting features **224** include keying features **228** for keyed mating with the mounting assembly **300**. For example, the keying features **228** may be ribs, tabs or other types of protrusions extending into the openings **226**. In other embodiments, the keying features **228** may be grooves, slots or other types of openings.

[0036] In an exemplary embodiment, the frame **202** includes support walls **230** extending from the rear **208**. The support walls **230** may support portions of the mounting assembly **300**. The support walls **230** may be aligned with the mounting features **224**. In the illustrated embodiment, four support walls **230** are provided near the four corners of the brick **220**. The support walls **230** may be used to separate or isolate the mounting assembly **300** from the cables **150**, such as to prevent damage to the cables and/or prevent obstruction of operation of the mounting assembly **300**.

[0037] In an exemplary embodiment, the mounting assembly **300** is operably coupled between the connector holder **200** and the front panel **114**. The mounting assembly **300** allows the connector holder **200**, and thus the cable connector(s) **130** to move relative to the front panel **114**. For example, the mounting assembly **300** allows the connector holder **200** and the cable connector **130** to move along the mating axis **126** (Z-direction), such as to accommodate overtravel of the mating connector assembly **104** during mating. The mounting assembly **300** allows the connector holder **200** and the cable connector **130** to move in one or more lateral floating directions perpendicular to the mating direction, such as in a horizontal floating direction (X-axis) and/or a vertical floating direction (Y-axis).

[0038] In an exemplary embodiment, the mounting assembly **300** includes one or more locating assemblies **302** configured to interface with the connector holder **200** at the corresponding mounting features **224**. Each locating assembly **302** of the mounting assembly **300** includes a shaft **310**, a biasing member **330** operably coupled between the connector holder **200** and the shaft **310**, and a support element **350** configured to be operably coupled between the shaft **310** and the front panel **114** of the cartridge **110** to support the shaft **310** relative to the front panel **114**.

[0039] The support element **350** may be a front support configured to be coupled to the front panel **114** of the cartridge **110**. The support element **350** is coupled to the shaft(s) **310**, such as using fasteners **352**, clips, latches, welding, or other securing means. In an exemplary embodiment, the support element **350** includes an inner surface **354** configured to be coupled to the front surface of the panel **114**. In an exemplary embodiment, the support element **350** includes an elongated plate, such as extending along the top and/or bottom and/or sides of the opening **120**. The support element **350** may be planar. The support element **350** may be coupled to multiple shafts **310**. The support element **350** may include notches **356** that receive the guideposts **122**. The support element **350** may have other shapes in alternative embodiments. For example, the support element **350** may be circular and configured to mate to a single shaft **310**.

[0040] The shaft **310** is configured to be coupled to the support element **350**. For example, the shaft **310** is configured to extend through a corresponding locating opening **121** in the front panel **114** to interface with the support element **350**. The front panel **114** may be captured between the support element **350** and the shaft **310**. The locating opening **121** may be at or near the port **120** in the front panel **114**, such as being open to the port **120**. In an exemplary embodiment, the shaft **310** is undersized relative to the locating opening **121** to allow a limited amount of floating movement in the locating opening **121** in a float direction (X-direction and/or Y-direction) perpendicular to the mating direction (Z direction) to allow the connector holder **200** and the cable connector **130** to move relative to the panel **114**. In an exemplary embodiment, the biasing member **330** forward biases the connector holder **200** and the cable connector **130**, such as to press the connector holder **200** toward the rear side of the panel **114**. The biasing member **330** is compressible in a rearward compression direction parallel to the mating direction (for example, along the Z axis) to allow the connector holder **200** and the cable connector **130** to move rearward relative to the panel **114**. As such, the mounting assembly **300** allows floating movement of the connector holder **200**, and thus the cable connector(s) **130**, in X/Y/Z directions to align the cable connectors **130** with the mating connectors **106** (FIG. **4**) during the mating process.

[0041] The shaft **310** extends between a first end **312** and a second end **314**. The first end **312** is at a front and configured to be coupled to the support element **350**. The second end **314** is at a rear and configured to be coupled to the biasing member **330**. The shaft **310** may be generally cylindrical. In an exemplary embodiment, the shaft **310** includes a head **320** and a stem **322** extending from the head **320** to the second end **314**. The stem **322** may be threaded. The head **320** has a larger diameter than the diameter of the stem **322**. In an exemplary embodiment, the shaft **310** includes a boss **324** extending from the head **320** opposite the stem **322**. The boss **324** is at the first end **312**. The boss **324** may be cylindrical. The boss **324** has a smaller diameter than the head **320**. In an exemplary embodiment, the boss **324** is configured to pass through the locating opening **121** to interface with the support element **350**. The support element **350** may be coupled to the boss **324** by the fastener **352**. For example, the boss **324** may include a threaded bore **326** that receives the fastener **352**.

[0042] The shaft **310** is configured to be coupled to the frame **202**. For example, the shaft **310** is received in the corresponding opening **226** in the frame **202**. For example, the stem **322** may pass through the opening **226** to the rear **208**, such as along the support wall **230**. The head **320** may be received in the opening **226**. In an exemplary embodiment, the head **320** includes a keying feature

328. The keying feature **328** is configured to interface with the keying feature **228**. In the illustrated embodiment, the keying feature is a slot or groove formed along the exterior of the head **320** that receives the rib defining the keying feature **228**. The keying features **228**, **328** may prevent rotation of the shaft **310** relative to the frame **202**. In an exemplary embodiment, the head **320** may bottom out against a shoulder **229** in the opening **226** to position the shaft **310** relative to the frame **202**. The biasing member **330** may forward bias the shoulder **229** against the rear side of the head **320**.

[0043] The biasing member **330** is configured to be coupled to the corresponding shaft **310**. For example, the stem **322** may pass through the biasing member **330**. In the illustrated embodiment, the biasing member **330** is a compression spring, such as a coil spring. However, other types of biasing members may be used in alternative embodiments. The biasing member **330** may include other types of springs, such as a leaf spring, a Belleville spring, a wave spring, a torsion spring. The biasing member **330** may include another type of compression element, such as a foam compression member, a rubber compression member, and the like.

[0044] The biasing member **330** extends between a first end **332** and a second end **334**. The first end **332** is configured to engage the frame **202**, such as at or near the rear **208** of the frame **202**. The first end **332** may be received in a pocket at the rear **208** to position and hold the first end **332**. In an exemplary embodiment, a nut **336** is provided at the second end **334**. The nut **336** may be threadably coupled to the shaft **310**, such as to the threaded end of the stem **322**, to secure the biasing member **330** to the shaft **310**. For example, the shaft **310** may pass through the biasing member **330** to interface with the nut **336**. The biasing member **330** directly engages the frame **202** to forward bias the connector holder **200** and the cable connector **130**. For example, the biasing member **330** presses forward against the nut **336** to forward bias the connector holder **200** and the cable connector **130** in a forward position for mating with the mating electrical connector **106**. Other securing means may be provided in alternative embodiments. The biasing member **330** and the nut **336** may be received in a trough or cradle in the support wall **230**.

[0045] In an exemplary embodiment, the frame **202** is configured to slide rearward along the stems **322** of the shafts **310**. For example, the biasing member **330** is compressible to allow the connector holder **200** and the cable connector(s) **130** to move in the rearward compression direction (Z direction). The shaft **310** guides movement of the frame **202** in the rearward compression direction. For example, the shafts **310** may limit or restrict movement of the frame **202** to movement parallel to the mating axis (Z-direction).

[0046] In an exemplary embodiment, the locating openings **121** are used to control floating movement of the connector assembly **102**, such as in the lateral floating direction(s) (for example, X direction and/or Y direction). In an exemplary embodiment, each locating opening **121** is oversized relative to the shaft **310**, such as being oversized relative to the boss **324**, which is received in the locating opening **121**. For example, clearance gaps **123** may be provided around the boss **324** to allow the boss **324** to move or float laterally within the locating opening **121**. In an exemplary embodiment, the locating opening **121** is horizontally oversized relative to the boss **324** allowing the boss **324** to move side-to-side within the locating opening **121** to allow a limited amount of lateral floating movement of the connector holder **200** and the cable connector(s) **130** relative to the front panel **114**. As such, the connector housing **132** is movable in the X-direction (for example, side-to-side) relative to the panel **114**. The amount of lateral movement may be limited to a confined amount by the edges of the panel defining the locating opening **121**. The clearance gap in the side-to-side direction may be approximately 3.0 mm in various embodiments. In an exemplary embodiment, the locating opening **121** is vertically oversized relative to the boss **324** allowing the boss **324** to move up-and-down within the locating opening **121** to allow a limited amount of lateral floating movement of the connector holder **200** and the cable connector(s) **130** relative to the front panel **114**. As such, the connector housing **132** is movable in the Y-direction (for example, up and down) relative to the panel **114**. The amount of lateral movement may be limited to a confined amount by the edges of the panel **114** defining the locating opening **121**. The clearance gap in the up-and-down direction may be approximately 3.0 mm in various embodiments.

[0047] FIG. **7** is a front perspective view of a portion of the communication system **100** showing one of the connector assemblies **102** in accordance with an exemplary embodiment coupled to the front panel **114** of the cartridge **110**. FIG. **8** is a front perspective, exploded view of the connector assembly **102** in accordance with an exemplary embodiment. The size and shapes of the components of the connector assembly **102** may be different than the embodiment shown in FIGS. **5** and **6**. The connector assembly **102** includes one or more of the cable connectors **130**, the connector holder **200** holding the cable connector(s) **130**, and the mounting assembly **300** for mounting the connector holder **200** and the cable connector(s) **130** to the front panel **114** of the cartridge **110**.

[0048] In the illustrated embodiment, the connector assembly **102** includes two of the cable connectors **130** arranged side-by-side. However, greater or fewer cable connectors **130** may be provided in alternative embodiments. The cable connector **130** includes the connector housing **132** holding the contact assemblies **140** in the cavity **134**. In the illustrated embodiment, a different number of rows and/or columns of the contact assemblies **140** are provided.

[0049] The connector holder **200** is sized and shaped differently to accommodate the different size/shape cable connector(s) **130**. The connector holder **200** is configured to be coupled to the cartridge **110**, via the mounting assembly **300**, to position each cable connector **130** relative to the cartridge **110**. In an exemplary embodiment, the cable connector **130** is movable relative to the connector holder **200** to position the cable connector **130** during mating with the mating electrical connector **106** (FIG. **4**). For example, the connector holder **200** and the cable connector **130** may be movable along the mating axis **126** (along the Z-axis) to accommodate overtravel of the mating connector assembly **104** during mating. In various embodiments, the connector holder **200** and the cable connector **130** may have a limited amount of floating movement relative to the front panel **114** to accommodate misalignment of the cable connector **130** relative to the mating electrical connector **106**. For example, the mounting assembly **300** allows movement of the connector holder **200** and the cable connector **130** relative to the front panel **114** along the X-axis and/or along the Y-axis to accommodate misalignment of the cable connector **130** relative to the mating electrical connector **106**. The mounting assembly **300** may limit or confine the amount of floating movement in the lateral floating directions X/Y. For example, the mounting assembly **300** may limit floating movement, such as to approximately 5.0 mm for mating tolerance. In various embodiments, the mounting assembly **300** may limit floating movement to approximately 3.0 mm for mating tolerance.

[0050] The connector holder **200** includes the frame **202** and the cable connector(s) **130** are received in the connector chamber **204** of the frame. In an exemplary embodiment, the frame **202** includes the upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and the second side frame member **216**. The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be plates or blocks. The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and the second side frame member **216** form the brick **220** used to hold the cable connector(s) **130**. The brick **220** is configured to be coupled to the cartridge **110** to hold the cable connector **130** relative to the cartridge **110**. In the illustrated embodiment, the mounting features **224** include openings **226** in the first side frame member **214** and/or the second side frame member **216**. In the illustrated embodiment, two openings **226** are provided along the sides of the brick **220**. The guideposts **122** extend forward from the frame **202**, such as from the first side frame member **214** and/or the second side frame member **216**. The guideposts **122** pass through openings in the front panel **114**.

[0051] The mounting assembly **300** includes the locating assemblies **302** configured to interface with the connector holder **200** at the corresponding mounting features **224**. The mounting assembly

300 is operably coupled between the connector holder **200** and the front panel **114**. The mounting assembly **300** allows the connector holder **200**, and thus the cable connector(s) **130**, to move relative to the front panel **114**. For example, the mounting assembly **300** allows the connector holder **200** and the cable connector **130** to move along the mating axis **126** (Z-direction), such as to accommodate overtravel of the mating connector assembly **104** during mating. The mounting assembly **300** allows the connector holder **200** and the cable connector **130** to move in one or more lateral floating directions perpendicular to the mating direction, such as in a horizontal floating direction (X-axis) and/or a vertical floating direction (Y-axis).

[0052] Each locating assembly **302** includes the shaft **310**, the biasing member **330** operably coupled between the connector holder **200** and the shaft **310**, and the support element **350** configured to be operably coupled between the shaft **310** and the front panel **114** of the cartridge **110** to support the shaft **310** relative to the front panel **114**. The support elements **350** may be sized and shaped differently than the embodiment shown in FIGS. **5** and **6**. For example, the support elements **350** are circular, such as being washers configured to be coupled to the front surface of the front panel **114** to secure the mounting assembly **300** to the front panel **114**.

[0053] The shaft **310** includes the head **320** at the first end **312** and the stem **322** at the second end **314**. The stem **322** is configured to be threadably coupled to the nut **336** to secure the biasing member **330** to the shaft **310**. The shaft **310** is configured to be coupled to the support element **350**. For example, the shaft **310** is configured to extend through a corresponding locating opening **121** in the front panel **114** to interface with the support element **350**. The front panel **114** may be captured between the support element **350** and the shaft **310**. The boss **324** at the front of the shaft is configured to pass through the locating opening **121** and the support element **350** is coupled to the boss **324** by the fastener **352**. The boss **324** of the shaft **310** is undersized relative to the locating opening **121** to allow a limited amount of floating movement in the locating opening **121** in a float direction (X-direction and/or Y-direction) perpendicular to the mating direction (Z direction) to allow the connector holder **200** and the cable connector **130** to move relative to the panel **114**. The biasing member **330** forward biases the connector holder **200** and the cable connector **130**, such as to press the connector holder **200** toward the rear side of the panel **114**. The biasing member **330** is compressible in a rearward compression direction parallel to the mating direction (for example, along the Z axis) to allow the connector holder **200** and the cable connector **130** to move rearward relative to the panel **114**. As such, the mounting assembly **300** allows floating movement of the connector holder **200**, and thus the cable connector(s) **130**, in X/Y/Z directions to align the cable connectors **130** with the mating connectors **106** (FIG. **4**) during the mating process.

[0054] When assembled, the shaft **310** is coupled to the frame **202** by loading the stem **322** of the shaft **310** in the corresponding opening **226** in the frame **202**. The head **320** is received in the opening **226** and the keying feature **328** interfaces with the keying feature **228** in the opening **226**. The stem **322** of the shaft passes through the biasing member **330** and is secured using the nut **336**. The biasing member **330** is captured between the frame **202** and the nut **336**. The biasing member **330** directly engages the frame **202** to forward bias the connector holder **200** and the cable connector **130** in a forward position for mating with the mating electrical connector **106**. In an exemplary embodiment, the frame **202** is configured to slide rearward along the stems **322** of the shafts **310** during mating with the mating connector **106**. The shaft **310** guides movement of the frame **202** in the rearward compression direction.

[0055] In an exemplary embodiment, the locating openings **121** are used to control floating movement of the connector assembly **102**, such as in the lateral floating direction(s) (for example, X direction and/or Y direction). In an exemplary embodiment, each locating opening **121** is oversized relative to the shaft **310**, such as being oversized relative to the boss **324**, which is received in the locating opening **121**. For example, the clearance gaps **123** are provided around the boss **324** to allow the boss **324** to move or float laterally within the locating opening **121** in the X-direction (for example, side-to-side) and/or in the Y-direction (for example, up and down) relative to the panel **114**. The amount of lateral movement may be limited to a confined amount by the edges of the panel **114** defining the locating opening **121**. The clearance gaps **123** may be approximately 3.0 mm in various embodiments.

[0056] FIG. **9** is a cross-sectional view of a portion of the communication system **100** showing the connector assembly **102** mounted to the front panel **114** of the cartridge **110** in accordance with an exemplary embodiment. The cable connector **130** is held relative to the cartridge **110** by the connector holder **200**. The connector holder **200** is held relative to the cartridge **110** by the mounting assembly **300**. For example, the shaft **310** is coupled to the front panel **114** by the support element **350** and the shaft **310** is coupled to the connector holder **200** by the biasing member **330**. The mating end of the cable connector **130** passes through the front panel **114** for mating with the mating electrical connector **106**.

[0057] When assembled, the shaft **310** passes through the opening **226** in the frame **202**. The head **320** bottoms out against the shoulder **229**. The boss **324** protrudes forward of the head **320** and is configured to interface with the support element **350**. The boss **324** passes through the locating opening **121** in the front panel **114** to interface with the support element **350**. In an exemplary embodiment, the boss **324** holds the support element **350** at a spaced apart location such that a clearance gap is provided between the head **320** and the support element **350** that receives the front panel **114**. The clearance gap may be wider than the thickness of the front panel **114** to allow free movement of the mounting assembly **300** relative to the front panel **114** in the floating direction(s). For example, the mounting assembly **300** does not bind on the front panel **114**, but rather is able to freely move in the X and Y directions within the clearance gap **123**.

[0058] During mating, the guidepost **122**, which protrudes forward of the panel **114** and the cable connector **130**, is configured to interface with the mating connector assembly **104** (shown in FIG. **4**). The guidepost **122** is used to generally align the mating connector assembly **104** with the cable connector **130**. If the cable connector **130** is misaligned (for example, vertically offset or horizontally offset), the mounting assembly **300** allows floating movement of the connector holder **200**, and thus the cable connector **130**, to align with the mating connector assembly **104**. For example, the connector holder **200** and the cable connector **130** are able to move in a floating direction, such as side-to-side (X direction) and/or up-and-down (Y direction), to accommodate the misalignment. During mating, the mating connector assembly **104** is plugged into the rack or chassis in a mating direction along the mating axis **126**. The mating ends of the mating electrical connector **106** is plugged into the connector housing **132** to electrically connect the signal contacts and the shields with the mating electrical connector **106**. Once fully mated, further loading of the mating connector assembly **104** in the mating direction causes compression of the mounting assembly **300**. For example, the biasing members **330** are compressed allowing the frame **202** to move in a rearward compression direction to accommodate the overloading of the mating connector assembly **104**.

[0059] The biasing members **330** are compressible in the rearward compression direction to allow the cable connector **130** to move to a rearward position, which prevents damage to the cable connector **130** and the mating electrical connector **106** during the mating process. In an exemplary embodiment, the biasing members **330** may be preloaded to a predetermined spring force. As such, the biasing members **330** are not compressed until the predetermined spring force is exceeded. The preload spring force may be greater than the mating force to fully mate the mating electrical connector **106** with the cable connector **130**. For example, the mating force may be defined by the force required to mate the signal contacts and the shields. In various embodiments, the mating force may be less than 20 pounds. The preload spring force may be set at 20 pounds to allow mating of the signal contacts and the shields without rearward movement of the cable connector **130**. The biasing members **330** compress only when the preload force is exceeded.

[0060] FIG. **10** is a perspective view of a portion of the communication system **100** showing one of the connector assemblies **102** in accordance with an exemplary embodiment. The connector assembly **102** is sized and shaped differently for mounting to the cartridge **110** at the side panels **116**, and/or the rear panel **118** rather than the front panel **114**. The mating end of the cable connector **130** passes through the opening **120** at the front panel **114**, but the connector assembly **102** is supported by the side panels **116**, and/or the rear panel **118** rather than the front panel **114**.

[0061] The connector assembly **102** includes one or more of the cable connectors **130**, the connector holder **200** holding the cable connector(s) **130**, and the mounting assembly **300** for mounting the connector holder **200** and the cable connector(s) **130** to the front panel **114** of the cartridge **110**. The mounting assembly **300** is configured to be mounted to the side panels **116** and/or the rear panel **118** rather than the front panel **114**.

[0062] FIG. **11** is a front perspective view of a portion of the communication system **100** showing one of the connector assemblies **102** in accordance with an exemplary embodiment. FIG. **12** is a front perspective, exploded view of a portion of the connector assembly **102** showing the cable connectors **130** and the connector holder **200** in accordance with an exemplary embodiment. FIG. **13** is a front perspective, exploded view of a portion of the connector assembly **102** showing the mounting assembly **300** in accordance with an exemplary embodiment.

[0063] In the illustrated embodiment, the connector assembly **102** includes two cable connectors **130** arranged side-by-side. However, greater or fewer cable connectors **130** may be provided in alternative embodiments. Each cable connector **130** includes the connector housing **132** holding the contact assemblies **140** in the cavity **134**. The contact assemblies **140** are arranged in a plurality of rows and columns. Each contact assembly **140** includes the signal contacts **142** electrically connected to the corresponding cable **150** and the shield **144** providing electrical shielding for the signal contacts **142**.

[0064] The connector holder **200** is used to hold the cable connector(s) **130** relative to the cartridge **110**. The connector holder **200** is configured to be coupled to the cartridge **110**, via the mounting assembly **300**, to position each cable connector **130** relative to the cartridge **110**. In an exemplary embodiment, the cable connector **130** is movable relative to the connector holder **200** to position the cable connector **130** during mating with the mating electrical connector **106** (FIG. **4**). For example, the connector holder **200** and the cable connector **130** may be movable along the mating axis **126** (along the Z-axis) to accommodate overtravel of the mating connector assembly **104** during mating. In various embodiments, the connector holder **200** and the cable connector **130** may have a limited amount of floating movement relative to the front panel **114** to accommodate misalignment of the cable connector **130** relative to the mating electrical connector **106**. For example, the mounting assembly **300** allows movement of the connector holder **200** and the cable connector **130** relative to the cartridge **110** along the X-axis and/or along the Y-axis to accommodate misalignment of the cable connector **130** relative to the mating electrical connector **106**. The mounting assembly **300** may limit or confine the amount of floating movement in the lateral floating directions X/Y. For example, the mounting assembly **300** may limit floating movement, such as to approximately 5.0 mm for mating tolerance. In various embodiments, the mounting assembly **300** may limit floating movement to approximately 3.0 mm for mating tolerance.

[0065] The connector holder **200** includes the frame **202** defining a connector chamber **204**. The cable connector(s) **130** are received in the connector chamber **204**. The frame **202** extends between the front **206** and the rear **208** of the connector holder **200**. The guideposts **122** extend forward from the front **206**, such as along the sides of the cable connectors **130**. Other locations are possible in alternative embodiments, such as above and/or below the cable connectors **130**.

[0066] In an exemplary embodiment, the frame **202** includes the upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and the second side frame member **216**. The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be plates or blocks. The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and the second side frame member

216 form the brick **220** used to hold the cable connector **130**. The brick **220** is configured to be coupled to the cartridge **110** to hold the cable connector **130** relative to the cartridge **110**.

[0067] In an exemplary embodiment, the mounting assembly **300** the locating assembly **302** configured to interface with the connector holder **200** at the corresponding mounting features **224**. The locating assembly **302** includes the shafts **310**, the biasing members **330** operably coupled between the connector holder **200** and the shafts **310**, a support element **370** configured to be operably coupled between the shaft **310** and the cartridge **110** to support the shafts **310** relative to the cartridge **110**, and sliders **390** operably coupled between the connector holder **200** and the support element **370**.

[0068] The support element **370** may be a rear support configured to be coupled to the cartridge **110** at or near the rear panel **118**. The support element **370** may be coupled to the side panels **116** and/or the rear panel **118**. The support element **370** is configured to be coupled to the side panels **116** by fasteners **372**. The support element **370** is coupled to the connector holder **200** through the shafts **310** and the biasing members **330**. In an exemplary embodiment, the support element **370** includes side walls **374**, **376** and an end wall **378** between the side walls **374**, **376**. For example, the support element **370** may be U-shaped. The end wall **378** is configured to extend along the rear panel **118**. The side walls **374**, **376** are configured to extend along the side panels **116**. The support element **370** may have other shapes in alternative embodiments. In various embodiments, the support element **370** may be a multi-piece support, such as including the side walls **374**, **376** without the end wall **378** therebetween.

[0069] Each shaft **310** is configured to be coupled to the connector holder **200** and the support element **370**. For example, the shafts **310** may be coupled to support tabs **380** extending from the side walls **374**, **376**. The sliders **390** may be positioned between the support tabs **380** and the frame **202**, such as the side frame members **214**, **216**. The shafts **310** may pass through locating openings **382** in the support tabs **380**. The support tabs **380** and the locating openings **382** may be fixed relative to the panels of the cartridge **110**; however, the shaft **310** may be movable relative to the support tabs **380** in the locating openings **382**. The shaft **310** is thus movable relative to the cartridge **110** to allow alignment of the cable connector **130** with the mating connector assembly **104**. In an exemplary embodiment, the shaft **310** is undersized relative to the locating openings **382** to allow a limited amount of floating movement in the locating openings **382** in a float direction (X-direction and/or Y-direction) perpendicular to the mating direction (Z direction) to allow the connector holder **200** and the cable connector **130** to move relative to the cartridge **110**. In an exemplary embodiment, the biasing member **330** forward biases the connector holder **200** and the cable connector **130** relative to the support element **370**. The biasing member **330** is compressible in a rearward compression direction parallel to the mating direction (for example, along the Z axis) to allow the connector holder **200** and the cable connector **130** to move rearward relative to the panel **114**. As such, the mounting assembly **300** allows floating movement of the connector holder **200**, and thus the cable connector(s) **130**, in X/Y/Z directions to align the cable connectors **130** with the mating connectors **106** (FIG. **4**) during the mating process.

[0070] The shaft **310** extends between a first end **312** and a second end **314**. The first end **312** is at a front and configured to be coupled to the frame **202**, such as to the side frame members **214**, **216**. The first end **312** may be threadably coupled to the frame **202**. The second end **314** is at a rear and is configured to be coupled to the support element **370**. The shaft **310** may be generally cylindrical. In an exemplary embodiment, the shaft **310** includes the head **320** and the stem **322** extending from the head **320** to the first end **312**. The stem **322** is configured to pass through the corresponding slider **390**. The stem **322** may pass through the locating openings **382** in the support tabs **380**. The head **320** is configured to be coupled to the rear surface of the support tabs **380**.

[0071] The biasing member **330** is configured to be coupled to the corresponding shaft **310**. For example, the stem **322** may pass through the biasing member **330**. The biasing members **330** are configured to be positioned between the sliders **390** and the support tabs **380**. The ends of the biasing members **330** are received in pockets in the rear end of the sliders **390**. The biasing member s**330** directly engage the sliders **390** to forward bias the sliders **390** toward the frame **202**. The sliders **390** may press the frame **202** in a forward biasing direction in a forward position for mating with the mating electrical connector **106**.

[0072] In an exemplary embodiment, the sliders **390** and the shafts **310** may slide rearward relative to the support tabs **380** of the support element **370**. For example, the biasing members **330** are compressible to allow the connector holder **200** and the cable connector(s) **130** to move in the rearward compression direction (Z direction). The shafts **310** guide movement of the frame **202** in the rearward compression direction. For example, the shafts **310** may limit or restrict movement of the frame **202** to movement parallel to the mating axis (Z-direction).

[0073] In an exemplary embodiment, the locating openings **382** are used to control floating movement of the connector assembly **102**, such as in the lateral floating direction(s) (for example, X direction and/or Y direction). In an exemplary embodiment, each locating opening **382** is oversized relative to the shaft **310**, such as being oversized relative to the stem **322**, which is received in the locating opening **382**. For example, clearance gaps **383** may be provided around the stem **322** to allow the stem **322** to move or float laterally within the locating opening **382**. In an exemplary embodiment, the locating opening **382** is horizontally oversized relative to the stem **322** allowing the stem **322** to move side-to-side within the locating opening **382** to allow a limited amount of lateral floating movement of the connector holder **200** and the cable connector(s) **130** relative to the support element **370** and thus relative to the cartridge **110**. As such, the connector housing **132** is movable in the X-direction (for example, side-to-side) relative to the cartridge **110**. The amount of lateral movement may be limited to a confined amount by the edges defining the locating opening **382**. The clearance gap in the side-to-side direction may be approximately 3.0 mm in various embodiments. In an exemplary embodiment, the locating opening **382** is vertically oversized relative to the stem **322** allowing the stem **322** to move up-and-down within the locating opening **382** to allow a limited amount of lateral floating movement of the connector holder **200** and the cable connector(s) **130** relative to the support element **370** and thus the cartridge **110**. As such, the connector housing **132** is movable in the Y-direction (for example, up and down) relative to the cartridge **110**. The amount of lateral movement may be limited to a confined amount by the edges defining the locating opening **382**. The clearance gap in the up-and-down direction may be approximately 3.0 mm in various embodiments.

[0074] In an exemplary embodiment, an anti-rotation pin **392** extends along the slider **390** between the frame **202** and the support element **370**. The anti-rotation pin **392** prevents rotation of the slide **390** during compression of the biasing member **330**.

[0075] It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope. Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase "means for" followed by a statement of function void of further structure.

Claims

1. A connector assembly comprising: a connector holder having a frame defining a connector chamber, the frame extending between a front and a rear of the connector holder, the frame including an upper frame member, a lower frame member, and first and second side frame members extending between the upper and lower frame members to define the connector chamber; a cable connector received in the connector chamber, the cable connector including a connector housing holding contact assemblies, each contact assembly including a signal contact and a cable terminated to the signal contact, the signal contact configured to be mated with a mating signal contact of a mating connector in a mating direction; and a mounting assembly for mounting the connector holder to a panel, the mounting assembly including a shaft, a biasing member operably coupled between the connector holder and the shaft, and a support element configured to be operably coupled between the shaft and the panel to support the shaft relative to the panel, the shaft configured to be received in a locating opening, wherein the shaft is undersized relative to the locating opening to allow a limited amount of floating movement in the locating opening in a float direction perpendicular to the mating direction to allow the connector holder and the cable connector to move relative to the panel, the biasing member forward biasing the cable connector and being compressible in a rearward compression direction parallel to the mating direction to allow the connector holder and the cable connector to move rearward relative to the panel.
2. The connector assembly of claim 1, wherein the shaft holds the support element spaced apart from the front of the frame to form a clearance gap forward of the frame, the clearance gap configured to receive the front panel.
3. The connector assembly of claim 2, wherein the clearance gap is wider than a thickness of the front panel such that the frame is spaced apart from the front panel.
4. The connector assembly of claim 1, wherein the support element is secured to the shaft by a fastener.
5. The connector assembly of claim 1, wherein the shaft is configured to move in at least two lateral float directions in the locating opening.
6. The connector assembly of claim 1, wherein the shaft includes a head and a stem extending from the head, the head being received in the frame, the frame configured to slide along the stem as the biasing member is compressed.
7. The connector assembly of claim 6, wherein the shaft includes a boss extending forward from the head, the boss passing through the locating opening, the support element engaging the boss.
8. The connector assembly of claim 7, wherein the boss is movable in the locating opening in the float direction.
9. The connector assembly of claim 1, wherein the frame includes a pocket at the rear, the pocket receiving the biasing member.
10. The connector assembly of claim 1, wherein the frame includes a front opening and a shoulder extending into the front opening, the shaft received in the front opening and engaging the shoulder to position the shaft relative to the frame.
11. The connector assembly of claim 1, wherein the mounting assembly is coupled to at least one of the upper frame member and the lower frame member.
12. The connector assembly of claim 1, wherein the mounting assembly is coupled to at least one of the first side frame member and the second side frame member.
13. The connector assembly of claim 1, wherein the mounting assembly includes a second shaft, a second biasing member operably coupled between the connector holder and the second shaft, and a second support element configured to be operably coupled between the second shaft and the panel

to support the second shaft relative to the panel, the second shaft configured to be received in a second locating opening in the panel, wherein the second shaft is undersized relative to the second locating opening to allow a limited amount of floating movement in the second locating opening in a float direction perpendicular to the mating direction, the second biasing member forward biasing the cable connector and being compressible in a rearward compression direction parallel to the mating direction.
14. The connector assembly of claim 1, further comprising a guide pin extending forward of the frame to guide mating with the mating connector.
15. The connector assembly of claim 1, wherein the support element includes a support tab, the locating opening provided in the support tab.
16. A connector assembly comprising: a connector holder having a frame defining a connector chamber, the frame extending between a front and a rear of the connector holder, the frame including an upper frame member, a lower frame member, and first and second side frame members extending between the upper and lower frame members to define the connector chamber; a first cable connector received in the connector chamber, the first cable connector including a first connector housing holding first contact assemblies, each first contact assembly including a first signal contact and a first cable terminated to the first signal contact, the first signal contact configured to be mated with a first mating signal contact of a first mating connector in a mating direction; a second cable connector received in the connector chamber, the second cable connector including a second connector housing holding second contact assemblies, each second contact assembly including a second signal contact and a second cable terminated to the second signal contact, the second signal contact configured to be mated with a second mating signal contact of a second mating connector in a mating direction; and a mounting assembly for mounting the connector holder to a panel, the mounting assembly including a shaft, a biasing member operably coupled between the connector holder and the shaft, and a support element configured to be operably coupled between the shaft and the panel to support the shaft relative to the panel, the shaft configured to be received in a locating opening, wherein the shaft is undersized relative to the locating opening to allow a limited amount of floating movement in the locating opening in a float direction perpendicular to the mating direction to allow the connector holder and the cable connector to move relative to the panel, the biasing member forward biasing the cable connector and being compressible in a rearward compression direction parallel to the mating direction to allow the connector holder and the cable connector to move rearward relative to the panel.
17. The connector assembly of claim 16, wherein the mounting assembly includes a second shaft, a second biasing member operably coupled between the connector holder and the second shaft, and a second support element configured to be operably coupled between the second shaft and the panel to support the second shaft relative to the panel, the second shaft configured to be received in a second locating opening in the panel, wherein the second shaft is undersized relative to the second locating opening to allow a limited amount of floating movement in the second locating opening in a float direction perpendicular to the mating direction, the second biasing member forward biasing the cable connector and being compressible in a rearward compression direction parallel to the mating direction.
18. The connector assembly of claim 16, wherein the shaft includes a head, a stem extending rearward from the head, and a boss extending forward from the head, the head being received in the frame, the stem holding the biasing member, the frame configured to slide along the stem as the biasing member is compressed, wherein the boss passes through the locating opening to interface with the support element, the boss being movable in the locating opening in the float direction.
19. A communication system comprising: a cartridge having panels forming a chamber, the panels including a front panel, the cartridge including ports in the front panel at a front of the cartridge; connector assemblies received in the ports for mating with mating connector assemblies, each connector assembly including a connector holder, a cable connector held by the connector holder,

and a mounting assembly for mounting the connector holder and the cable connector to the cartridge, each connector holder having a frame defining a connector chamber, the frame extending between a front and a rear of the connector holder, the frame including an upper frame member, a lower frame member, and first and second side frame members extending between the upper and lower frame members to define the connector chamber; each cable connector received in the corresponding connector chamber, each cable connector including a connector housing holding contact assemblies, each contact assembly including a signal contact and a cable terminated to the signal contact, the signal contact configured to be mated with a mating signal contact of a mating connector in a mating direction; and each mounting assembly including a shaft, a biasing member operably coupled between the connector holder and the shaft, and a support element configured to be operably coupled between the shaft and the panels of the cartridge to support the shaft relative to the panels, the shaft configured to be received in a locating opening, wherein the shaft is undersized relative to the locating opening to allow a limited amount of floating movement in the locating opening in a float direction perpendicular to the mating direction to allow the connector holder and the cable connector to move relative to the front panel, the biasing member forward biasing the cable connector and being compressible in a rearward compression direction parallel to the mating direction to allow the connector holder and the cable connector to move rearward relative to the front panel.

20. The communication system of claim 19, wherein the shaft includes a head, a stem extending rearward from the head, and a boss extending forward from the head, the head being received in the frame, the stem holding the biasing member, the frame configured to slide along the stem as the biasing member is compressed, wherein the boss passes through the locating opening to interface with the support element, the boss being movable in the locating opening in the float direction.

21. The communication system of claim 19, further comprising a guide pin coupled to the frame and extending forward of the front panel, the guide pin configured to engage a guide module of the mating connector assembly to guide mating of the mating connectors with the cable connectors.